ClaimAI

Project summary

Author: Ali Bidanov, 3rd-year student, International Information Technology University (34/1 Manas St., Almaty), alibig860@gmail.com.

Goal: to cut the time and improve the consistency of card dispute handling with a large language model while keeping the human operator in control.

Relevance

According to Mastercard, global chargeback volume is expected to grow from 261 million in 2025 to 324 million in 2028, and dispute value from $33.8 billion to $41.7 billion. Handling one dispute costs an issuing bank roughly $9–10. In Kazakhstan, more than 130,000 fraud cases were recorded from the start of 2025 to March 2026 (National Bank). Manual review does not scale and is hard to audit.

Description

ClaimAI takes a customer complaint and the case transactions, identifies the dispute type (fraud, ATM cash not dispensed, duplicate charge, transaction series) and produces a decision: verdict, risk and confidence scores from 0 to 100, reasoning, a list of actions and a customer letter. The operator confirms the decision and the system records it in an audit log.

What is built

- React web interface and FastAPI backend with a PostgreSQL database.
- LLM decision engine (OpenAI API) with a strict JSON response format, validation and normalization.
- Audit log: operator ID, timestamp, decision and reasoning.
- Four synthetic demo cases; deployment via Docker, Vercel and Render.

Novelty

A combination of an LLM, a structured decision and an operator-confirmed audit trail for Russian-language complaints and tenge-denominated transactions.

Next steps

- Knowledge base of Visa and Mastercard rules (RAG) with rule citations.
- Accuracy evaluation on labelled cases and masking of personal data.
- Pilot at a bank or processor; option of an on-premise model.

Links: repository [GitHub link], demo [link to the working prototype].